

Caregiver Failed Extubation Rate and Patient Mortality:

Working Document — June 2026

Zack Treisman

This is a working document summarizing analyses completed to date and addressing questions raised by Andrew and Ryan. Blue boxes are remaining action items. All analyses use MIMIC-IV v3.1 (Beth Israel Deaconess Medical Center, 2008–2022). The primary outcome is **12-month mortality** throughout; all results from prior versions that used 12-month survival have been converted to the mortality framing for consistency with hospital and 30-day mortality secondary endpoints.

1 Cohort and Data

1.1 Invasive vs. non-invasive ventilation

There is an indicator variable in MIMIC-IV distinguishing invasive (endotracheal) intubation from non-invasive ventilation (CPAP/BiPAP). I confirmed that the pipeline filters to `InvasiveVent` only. The other categories are `SupplementalOxygen`, `NonInvasiveVent`, `HFNC`, `Tracheostomy`.

1.2 Cohort sizes and covariate comparison

Tables 1–3 summarize cohort flow and characteristics. The primary analysis cohort (`analysis_base`) includes 3,426 patients with explicit extubation events, caregiver volume ≥ 20 at the unit level, and standard ICU unit labels. The `patient_cohort` ($N = 12,662$) is used for descriptive analyses of the full population of invasively ventilated patients with extubation events, including those without a caregiver link and those in non-ICU units.

When the extubation event is inferred from the ventilation record rather than explicitly documented, there is no caregiver link and the patient is excluded from the primary analysis. These patients are included in the `patient_cohort` for descriptive purposes but not in any caregiver-level analyses. An additional complication with these patients is that vent duration is highly uncertain. Attempts to impute caregiver assignment for inferred events have not yet yielded a reliable method; this remains an open question for future work.

A note on caregiver attribution: `caregiver_id` in MIMIC-IV records the *documenting* user, not necessarily the provider who performed the extubation. This is a known limitation of the dataset.

Non-ICU units excluded: Four unit labels were identified as non-standard settings and excluded from the primary analysis: PACU ($N = 19$), generic “ICU” ($N = 12$), Surgery/Vascular/Intermediate ($N = 76$, $\approx 50\%$ COVID diagnoses, vent hours $> 220h$, likely COVID-era overflow ward), and Medicine ($N = 11$). These patients are retained in `patient_cohort` but not in the primary analysis cohort.

Stage	N patients	Lost
InvasiveVent ≥ 4 h in MIMIC-IV v3.1	27,729	
of whom: explicit extubation documented	18,958	
of whom: extubation inferred from vent record	8,771	
<i>Filters applied to both groups:</i>		
Died in hospital without extubation		−6,050
Transitioned to tracheostomy		−1,523
No computable ventilation duration		−8,068
Incomplete covariates or non-ICU unit		−68
patient_cohort (last extubation per patient)	12,662	
Inferred events (no caregiver link)		−7,826
Caregiver total volume < 20 or non-standard unit		−810
explicit_extubations (caregiver.n ≥ 20)	4,026	
Unique caregivers	166	
ICU units	8	
Caregiver unit-specific volume < 20		−600
analysis_base (primary analysis cohort)	3,426	
CVICU	1,435	
Medical (MICU + MICU/SICU)	1,001	
Surgical/Trauma (SICU + TSICU)	809	
Other units (descriptive only)	181	

Table 1: Cohort flow from all invasively ventilated patients in MIMIC-IV to the primary analysis cohort. The **patient_cohort** ($N = 12,662$) is used for descriptive analyses. The **explicit_extubations** cohort ($N = 4,026$) requires a genuine caregiver assignment and a caregiver total volume ≥ 20 . The **analysis_base** ($N = 3,426$) additionally requires caregiver unit-specific volume ≥ 20 , which is the denominator used to compute **caregiver_fe_rate**. See Table 2 for the statistical rationale and caregiver-count cost at each threshold.

TODO: CONSORT diagram: Create a flow diagram visualizing the cohort flow from all invasively ventilated patients to the final analysis cohort, with counts and reasons for exclusion at each step.

1.3 Terminal extubation

The current analysis uses death within 3 days of extubation as the criterion for a likely comfort extubation, flagging 175 patients (4.3% of the analytical cohort). A sensitivity analysis excluding these cases is reported in Section 4.5.

Andrew’s note on code status: Death within 3 days is a rough proxy — it captures patients extubated with survival intent who had unexpected deterioration, not only those extubated as comfort care. A more specific marker would be a documented change in code status (DNR/DNI order) in the 24–48h preceding extubation, which is available in MIMIC-IV **chartevents**. This query has not yet been run; it is a priority for the next revision.

	$n \geq 5$	$n \geq 10$	$n \geq 20$	$n \geq 32$
<i>Statistical reliability (true FE rate = 5%)</i>				
$P(\text{observe } 0\% \text{ FE}) = 0.95^n$	77.4%	59.9%	35.8%	19.4%
$P(\text{observe } \geq 1 \text{ failure}) = 1 - 0.95^n$	22.6%	40.1%	64.2%	80.6%
<i>Eligible caregivers per unit (<code>caregiver_n</code> > 0)</i>				
CVICU	141	136	102	71
MICU	107	87	47	33
TSICU	102	89	48	31
SICU	97	73	40	28
MICU/SICU	67	55	32	21
CCU	53	40	26	15
Neuro SICU	25	17	4	0
Neuro Intermed.	19	9	1	0
Neuro Stepdown	10	1	0	0

Table 2: Statistical reliability of the caregiver FE rate label and caregiver retention at each unit-specific volume threshold. Top rows: probability of observing zero failures (falsely appearing 0% FE) vs. at least one failure for a caregiver whose true underlying rate is 5%. Bottom rows: number of eligible caregivers per unit at each threshold. The $n \geq 32$ threshold achieves 80% power to detect at least one failure but eliminates the Neuro units and reduces the medical/surgical groups by 30–40%. The analysis uses $n \geq 20$ as a practical compromise retaining all six primary units with ≥ 20 caregivers each. Neuro units are excluded from the primary MSM analysis regardless of threshold.

	All patients	<code>explicit_extubations</code>	<code>analysis_base</code>
N	12,662	4,026	3,426
Age, mean (SD)	62.7 (16.2)	63.1 (15.6)	63.3 (15.6)
Female, %	40.7	38.0	38.0
Charlson, mean (SD)	5.57 (3.25)	5.46 (3.23)	5.46 (3.23)
SOFA, mean (SD)	6.91 (3.86)	6.47 (3.52)	6.34 (3.45)
Norepinephrine, mean (SD)	0.17 (0.23)	0.15 (0.19)	0.14 (0.19)
Vent hours, mean (SD)	94.3 (134.1)	64.3 (95.1)	61.0 (91.8)
Medical-respiratory intubation, %	74.7	64.3	64.8
Circulatory primary diagnosis, %	32.9	42.6	42.8
12-month mortality, %	45.2	32.1	30.0
Hospital mortality, %	27.3	12.0	11.9

Table 3: Comparison of the full patient cohort vs. `explicit_extubations` (explicit events with `caregiver_n` ≥ 20 , standard ICU unit) vs. the primary analysis cohort `analysis_base` (additional `caregiver_unit_n` ≥ 20). The analytical cohorts are systematically less sick than the full `patient_cohort`: lower SOFA, shorter ventilation, lower mortality, reflecting selection toward patients with better-documented, shorter ICU stays. The two analytical cohorts are similar to each other; the unit-volume threshold (~ 600 patients) causes minimal further selection.

TODO: code status query: Run a query to identify patients with a documented change in code status to DNR/DNI in the 24–48h preceding extubation. Use this to refine the comfort extubation sensitivity analysis.

2 ICU Unit Breakdown

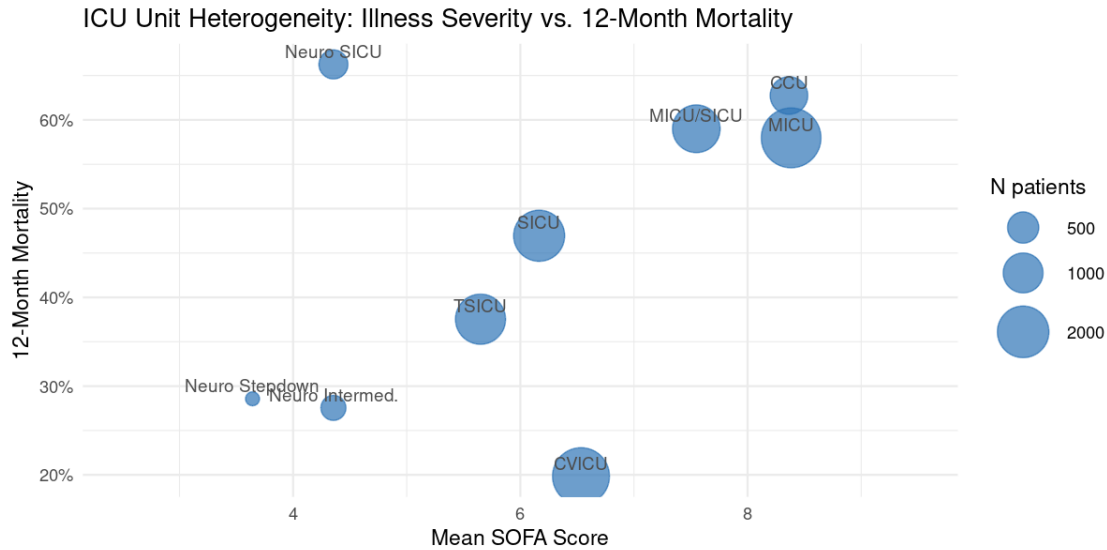


Figure 1: ICU unit heterogeneity: mean SOFA score vs. 12-month mortality. Bubble size proportional to patient count.

Unit	N	Age mean (SD)	SOFA mean (SD)	Charlson mean (SD)	Vent h median [IQR]	12-mo mortality %
CVICU	2,585	66.5 (12.5)	6.5 (3.3)	5.4 (2.9)	14 [6–45]	19.8
TSICU	1,854	58.7 (19.2)	5.7 (3.5)	4.3 (3.3)	44 [15–122]	37.5
SICU	1,947	62.7 (15.9)	6.2 (4.0)	5.6 (3.0)	42 [17–123]	46.9
MICU	2,915	61.3 (16.5)	8.4 (4.0)	6.0 (3.3)	64 [23–154]	58.0
MICU/SICU	1,622	61.8 (16.9)	7.6 (3.8)	6.1 (3.5)	54 [18–142]	59.0
CCU	851	67.6 (13.9)	8.4 (3.5)	7.0 (2.9)	50 [20–114]	62.7
Neuro SICU	403	64.7 (16.3)	4.4 (3.0)	5.7 (3.2)	33 [15–82]	66.3
Neuro Intermed.	265	58.9 (17.0)	4.4 (2.9)	4.9 (3.0)	35 [17–81]	27.5
Neuro Stepdown	98	58.6 (16.3)	3.6 (2.2)	4.4 (3.0)	25 [12–66]	28.6

Table 4: Patient profiles by ICU unit (`patient_cohort`, N=12,662; explicit + inferred events). Units ordered by 12-month mortality (ascending). Neuro units listed last. See Table 6 for unit-specific FE rates in the analytical cohort. There is no separate CTICU label in MIMIC-IV. The cardiac surgery recovery unit appears exclusively as “Cardiac Vascular Intensive Care Unit (CVICU).”

2.1 Unit clinical profiles

The eight recognized ICU units in the analytical cohort fall into four clinically distinct groups:

Cardiac Vascular ICU (CVICU, $N = 2,585$ in `patient_cohort`). Oldest patients (mean 66.5y), 84% circulatory diagnoses, short ventilation (55h mean), lowest FE rate (1.8%), best outcomes (13.7% 12-month mortality in `analysis_base`). Extubation is a near-routine step in surgical recovery. **Andrew’s note on CVICU vs. CTICU:** There is no separate “Cardiothoracic ICU” label in MIMIC-IV; all cardiac surgery and cardiac medical admissions appear under the single CVICU label. Among CVICU patients in the analytical cohort, 84.2% have a circulatory primary

ICD chapter and 45.2% have medical-non-respiratory intubation type (a proxy for cardiac surgery recovery). The remainder are likely post-MI and other non-surgical cardiac events.

Medical critical care (MICU + MICU/SICU, $N = 4,537$). Highest SOFA (8.4 and 7.6), longest ventilation (118h and 111h), 88%+ medical-respiratory intubation. MICU dominated by infectious (28%) and respiratory (20%) diagnoses. 12-month mortality approximately 48–56%.

Surgical and trauma (SICU + TSICU, $N = 3,801$). TSICU: youngest (58.7y), lowest Charlson (4.3), 46% injury/poisoning diagnoses. SICU: more heterogeneous (17.9% digestive, 15.5% injury, 11.2% neoplasms).

Cardiac care (CCU, $N = 851$). Oldest alongside CVICU (67.6y), highest Charlson (7.0), 64% circulatory diagnoses. 56.2% 12-month mortality despite “only” 40.8% hospital mortality — substantial post-discharge mortality. Excluded from the primary MSM analysis due to insufficient caregiver volume with the ≥ 20 unit-extubation threshold (2 eligible caregivers).

Neuro units ($N = 766$). **Andrew’s note:** The “Neuro Surgical ICU (Neuro SICU)” label in MIMIC-IV is used for patients with primary neurological diagnoses including both neurosurgical and medical neurology admissions (13.4% neurologic, 18.1% injury/poisoning, 51.4% circulatory — the last group perhaps representing aneurysm and AVM cases). It is not a pure neurosurgical unit. Neuro Intermediate and Stepdown have lower mortality (28% and 25%) and are step-down rather than full ICU settings; all three are excluded from the primary analysis due to small caregiver counts.

3 Caregiver Profiles

Table 5 summarizes the caregiver cohort flow. Of 131 unique caregivers in the primary analysis cohort, 207 caregiver–unit group assignments exist (CVICU + Medical + Surgical/Trauma = 102 + 52 + 53), meaning 76 caregivers provide care across multiple primary analysis groups and are counted independently within each group with their unit-specific FE rate.

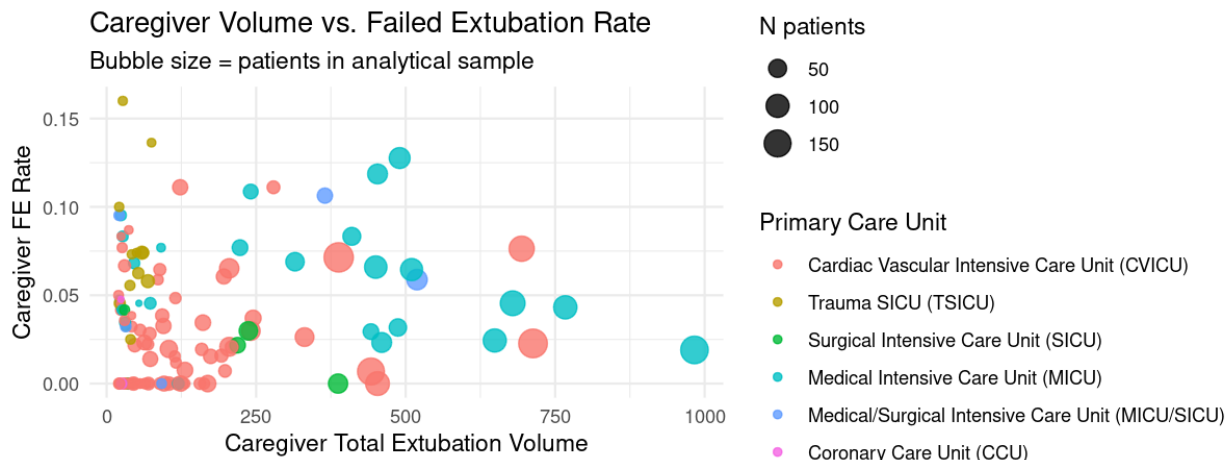


Figure 2: Caregiver total extubation volume vs. failed extubation rate. Bubble size = patients in analytical sample.

Stage	N caregivers	Lost
Caregivers in patient_cohort	570	
(all have explicit events; inferred events carry no caregiver link)		
Total extubation volume < 20 (<code>caregiver_n</code> < 20)		−404
Caregivers with volume ≥ 20	166	
Unit-specific volume < 20 in all units		−32
Caregivers in analysis_base (<code>caregiver_unit_n</code> ≥ 20)	134	
In “Other” ICU units only (CCU, Neuro, non-standard)		−3
Caregivers in primary analysis groups	131	
CVICU (unit-specific)	102	
Medical (MICU + MICU/SICU, unit-specific)	52	
Surgical/Trauma (SICU + TSICU, unit-specific)	53	
<i>Total caregiver–unit assignments</i>	<i>207</i>	

Table 5: Caregiver cohort flow. The three per-group caregiver counts ($102 + 52 + 53 = 207$) exceed the 131 unique caregivers because 76 caregivers work across multiple primary analysis groups and are counted independently in each with their unit-specific FE rate. Caregiver profiles by primary unit are shown in Table 6.

	CVICU	MICU	TSICU	SICU	MICU/SICU	CCU
N caregivers	79	26	13	6	7	3
<i>FE rate (unit-specific)</i>						
Median	1.4%	4.5%	7.3%	2.6%	4.8%	0%
IQR	0–3.6%	3.2–7.7%	5.6–7.4%	0.5–3.2%	1.6–7.7%	0–2.4%
Max	11.1%	12.8%	16.0%	4.2%	10.6%	4.8%
<i>Volume (total extubations)</i>						
Median	73	278	42	128	32	24
IQR	39–158	48–480	34–59	33–232	24–228	24–24
Max	713	983	75	387	519	25
<i>Patients in analytical sample</i>						
Median	11	32	6	20	7	6
12-mo mortality, mean	13.8%	39.6%	23.5%	34.6%	42.1%	58.3%

Table 6: Caregiver profiles by primary ICU unit (`caregiver_unit_n` ≥ 20 , consistent with `analysis_base`). Volume = total explicit extubation events in MIMIC-IV. CVICU has the largest number of eligible caregivers (79) despite lower individual volumes, reflecting a large cardiac surgery program. Median FE rate is 1.4% for CVICU caregivers; most with 0% rate are low-volume providers (see Section 4.1). MICU caregivers have the highest volume (median 278). CCU has only 3 caregivers meeting the threshold and is excluded from the primary MSM analysis.

4 Causal Analysis: Continuous Treatment MSM

4.1 Why we moved away from binary PSM

The original analysis compared patients of top-quartile FE rate caregivers (Q4) to bottom-quartile (Q1) caregivers. Andrew’s review identified two fundamental problems.

The 0%-FE stratum is driven by low volume. In CVICU, the bottom quartile consists entirely of caregivers who have had zero failed extubations in their career ($Q25 = 0\%$ at this unit). A characterization analysis (Figure 8) reveals that this group is dominated by *low-volume* providers who have not accumulated enough extubations to have had a failure yet. Among the 73 CVICU caregivers with a 0% FE rate, the median number of patients in the analytical sample is just 4 [IQR 2–6]. By contrast, providers with non-zero FE rates have median analytical-sample volumes of 16–22 patients. The probability of observing zero failures by chance (assuming a true underlying rate of 5%) is $(0.95)^4 \approx 81\%$ with four observations. The 0%-FE label is largely statistical noise for these low-volume providers.

The propensity score has a weak conceptual basis here. The propensity score models $P(\text{assigned to high-FE caregiver} \mid \text{patient covariates})$. As Andrew correctly noted, within an ICU, patient-to-caregiver assignment is largely determined by staffing rosters, not patient characteristics. Our analysis confirms this empirically: patient covariates explain only 4–7% of variance in caregiver FE rate in all three unit groups (denominator $R^2 = 0.057, 0.042, 0.069$ for CVICU, Medical, and Surgical/Trauma). The IPW weights are therefore close to 1 for most patients, meaning the weighting is performing almost no confound correction. This is consistent with near-random assignment.

4.2 Method: stabilized density weights

We replaced the binary PSM with a marginal structural model (MSM) for continuous treatment (Robins et al. 2000; Hernán & Robins 2020, §12). The exposure is `caregiver_fe_rate` as a continuous variable; all patients are included regardless of their caregiver’s quartile. Stabilized Hirano–Imbens density weights are:

$$sw_i = \frac{f(T_i)}{f(T_i \mid X_i)}$$

where $T = \text{caregiver_fe_rate}$, $X = \text{patient covariates}$ (age, sex, Charlson, SOFA, norepinephrine, ventilation hours, intubation type, 21 CCSR diagnosis flags), and f denotes a probability density function (PDF). $f(T_i)$ is the *marginal* density of the treatment evaluated at patient i ’s observed FE rate; $f(T_i \mid X_i)$ is the *conditional* density of treatment given patient covariates. Both are estimated as normal densities from OLS: fitting $T_i = \alpha + X_i\beta + \varepsilon_i$ gives conditional fitted values $\hat{T}_i$ and residual variance $\hat{\sigma}_{T|X}^2$, so

$$f(T_i \mid X_i) = \frac{\phi\left(\frac{T_i - \hat{T}_i}{\hat{\sigma}_{T|X}}\right)}{\hat{\sigma}_{T|X}}, \quad f(T_i) = \frac{\phi\left(\frac{T_i - \bar{T}}{\hat{\sigma}_T}\right)}{\hat{\sigma}_T},$$

where ϕ is the standard normal PDF and $\bar{T}$, $\hat{\sigma}_T$ are the marginal mean and standard deviation of T . The ratio upweights patients whose observed FE rate was unlikely given their covariates and downweights those for whom it was expected, recreating a pseudo-population in which covariates are balanced across FE rate levels. Weights are trimmed at the 99th percentile. Primary inference

uses cluster-robust standard errors by `caregiver_id`. The effect size is reported as the odds ratio per 5 percentage-point increase in caregiver FE rate (approximately the CVICU IQR).

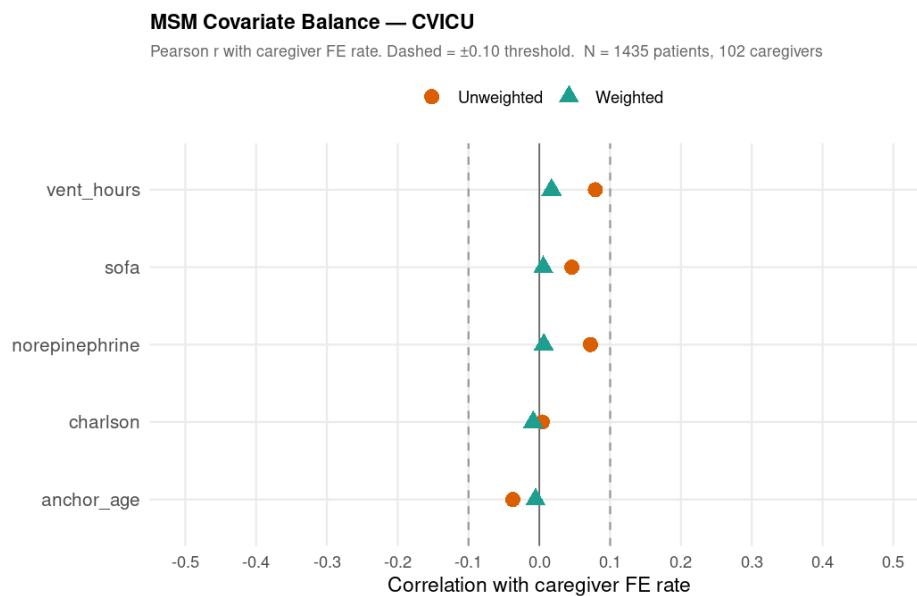


Figure 3: MSM covariate balance — CVICU. Pearson correlation of each covariate with caregiver FE rate before (orange) and after (teal) IPW weighting. Dashed lines = ± 0.10 threshold. All covariates fall within the threshold after weighting.

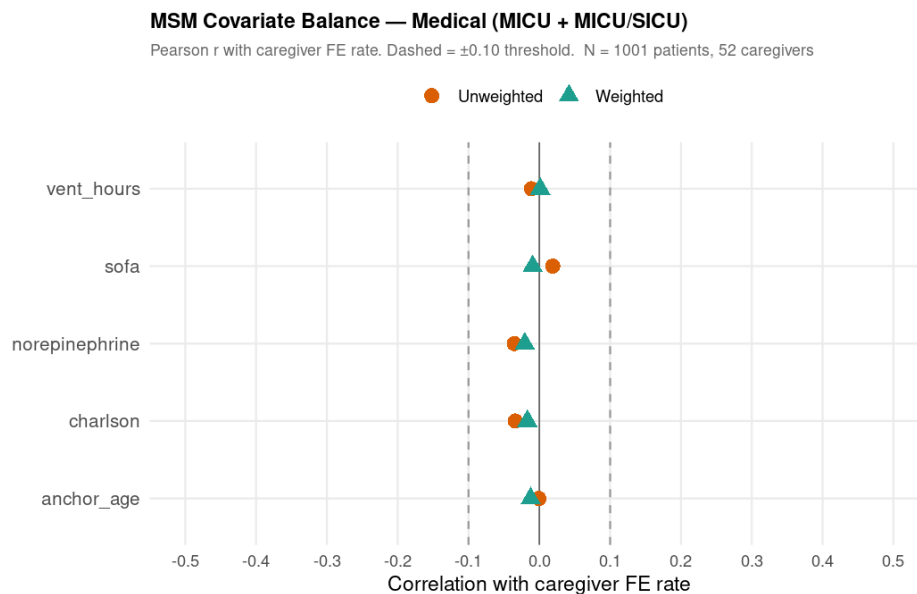


Figure 4: MSM covariate balance — Medical (MICU + MICU/SICU). Pearson correlation of each covariate with caregiver FE rate before (orange) and after (teal) IPW weighting. Dashed lines = ± 0.10 threshold.

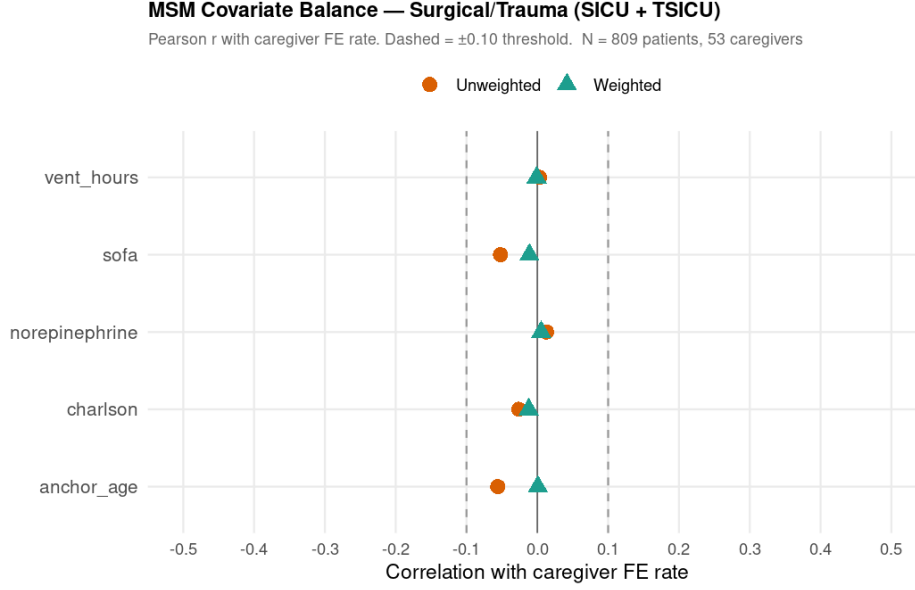


Figure 5: MSM covariate balance — Surgical/Trauma (SICU + TSICU). Pearson correlation of each covariate with caregiver FE rate before (orange) and after (teal) IPW weighting. Dashed lines = ± 0.10 threshold.

Group	N	Caregivers	Denom R^2	OR per 5pp	95% CI	p
CVICU	1,435	102	0.057	1.432	0.931–2.200	0.102
Medical	1,001	52	0.042	0.898	0.750–1.074	0.239
Surgical/Trauma	809	53	0.069	0.893	0.627–1.270	0.529

Table 7: MSM results by unit group. Outcome: 12-month mortality. OR > 1 indicates higher mortality for patients of higher-FE-rate caregivers. Denom R^2 = variance in caregiver FE rate explained by patient covariates in the denominator model; low values confirm near-random caregiver assignment. Cluster-robust SEs by `caregiver_id`.

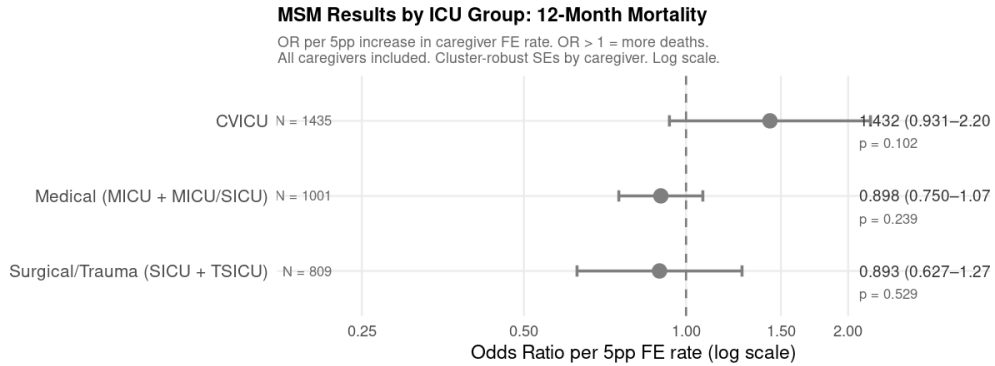


Figure 6: MSM results by ICU group: 12-month mortality odds ratio per 5 percentage-point increase in caregiver FE rate. OR > 1 = more deaths with higher FE rate. All groups are null ($p > 0.05$). Cluster-robust SEs by caregiver.

4.3 Results

The CVICU point estimate (OR = 1.432) is directionally consistent with the prior PSM finding (OR = 1.72 on the mortality scale), but does not reach significance ($p = 0.102$). The denominator $R^2 = 0.057$ confirms that patient covariates explain very little of caregiver FE rate variance, and the effective sample size after weighting (1,392 of 1,435) shows the weights are barely adjusting for confounding. Medical and Surgical/Trauma are null across all methods.

4.4 Secondary outcomes

Group	Hosp. mortality OR	p	30-day mortality OR	p
CVICU	1.408 (0.712–2.783)	0.325	2.144 (0.788–5.829)	0.135
Medical	1.120 (0.917–1.367)	0.267	1.029 (0.807–1.313)	0.817
Surgical/Trauma	0.856 (0.530–1.381)	0.523	0.960 (0.636–1.447)	0.844

Table 8: Secondary mortality outcomes by ICU group. OR > 1 = more deaths with higher FE rate. All three mortality endpoints are now in the same direction (OR > 1 = worse); CVICU shows a consistent positive direction across all three endpoints but does not reach significance at any. The Surgical/Trauma 30-day signal seen in the prior PSM analysis does not survive in the MSM.

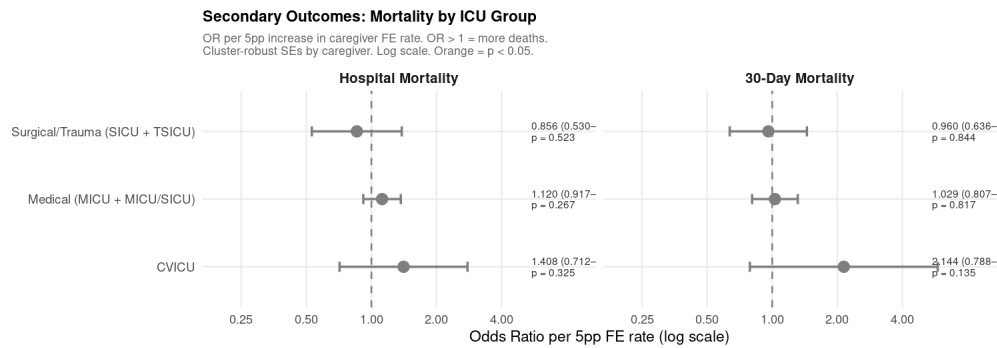


Figure 7: MSM secondary outcomes by ICU group: hospital and 30-day mortality. OR > 1 = more deaths with higher FE rate. All results are non-significant.

4.5 Comfort extubation sensitivity

Excluding likely comfort extubations (in-hospital death within 3 days; $N = 10$ in CVICU, 0.7%):

	MSM OR per 5pp	p
CVICU (all)	1.432	0.102
CVICU (comfort excluded)	1.398	0.122

The CVICU direction is stable and the result is not driven by comfort-care confounding.

4.6 Volume confounding and the 0%-FE stratum

The caregiver characterization analysis (Table 9) explains why the 0%-FE stratum shows better patient outcomes.

FE rate stratum	N (pts)	N (caregivers)	Med vol	Max vol	Vent h (med)	Mortality
0% (exact)	313	36	6 [4–10]	45	8 [5–19]	9.9%
>0–median (1.96%)	569	22	16 [9–26]	149	6 [4–19]	13.7%
median–Q75 (2.99%)	273	14	16 [12–22]	56	12 [5–37]	15.8%
>Q75	280	30	8 [4–12]	29	15 [5–36]	15.7%

Table 9: CVICU patient and caregiver characteristics by FE rate stratum (**analysis_base**, **caregiver_unit_n** ≥ 20). Med vol = median patients in analytical sample per caregiver [IQR]. The 0%-FE stratum has the lowest caregiver volumes and the shortest ventilation times — both consistent with a simpler case mix — rather than any identifiable superiority in extubation skill. SOFA, Charlson, and age are similar across all strata.

The 0%-FE caregivers have median volume of 6 patients [IQR 4–10] in the analytical sample, and their patients have shorter ventilation times (8h median) than all other strata. This profile is consistent with occasional-coverage providers seeing straightforward post-surgical cases who are easy to extubate. The caregiver volume and case mix, not the FE rate per se, likely explain the mortality difference in this stratum.

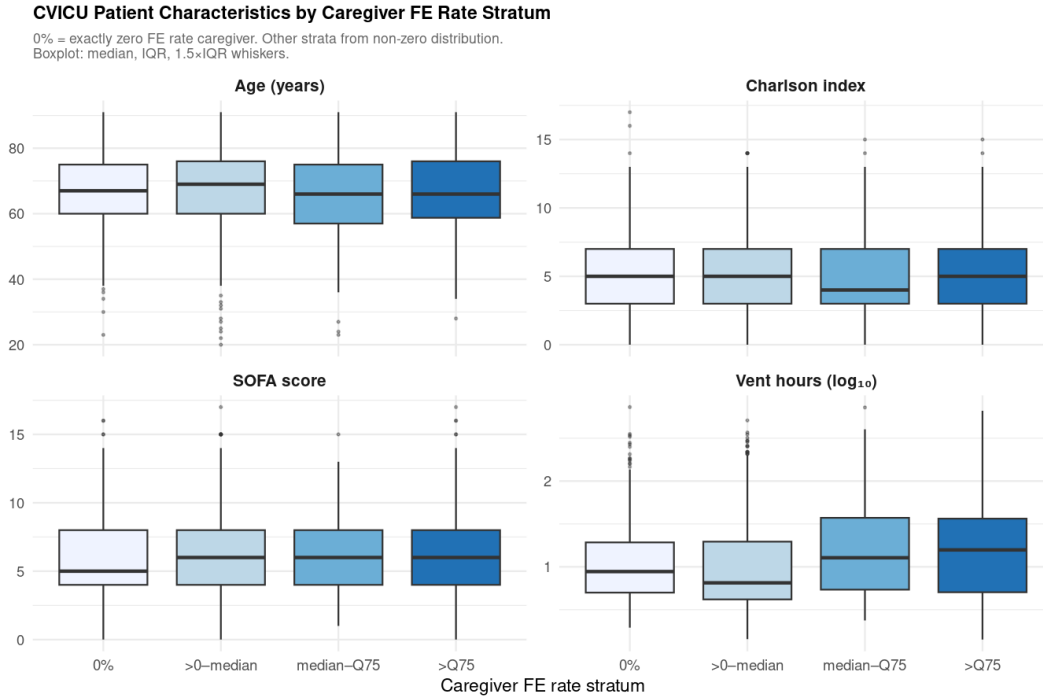


Figure 8: CVICU patient characteristics by caregiver FE rate stratum. SOFA, age, and Charlson are similar across strata. Ventilation hours increase monotonically with FE rate, consistent with higher FE rate caregivers managing more complex, longer-ventilated patients. Log scale on vent hours.

4.7 Caregiver-level partial correlations

At the caregiver level (one observation per caregiver, outcome = mean 12-month patient mortality), the partial correlation between FE rate and mortality controlling for caregiver volume is:

Cohort	N caregivers	r (FE rate $\sim$ mortality)	p
analysis_base (caregiver_unit_n $\geq$ 20)	134	+0.216	0.012
+ analytic patients $\geq$ 10	70	+0.432	< 0.001
+ analytic patients $\geq$ 20	42	+0.485	0.001

The positive sign confirms the direction: higher FE rate caregivers have patients with worse mortality, controlling for caregiver volume. The association strengthens when restricted to caregivers with larger analytical samples, suggesting it is not noise from small denominators. Volume also carries independent signal when controlling for FE rate ($r = +0.282$, $p = 0.001$).

The CVICU unit-stratified partial correlation is $r = +0.261$, $p = 0.021$ ($N = 79$ caregivers) — again specific to CVICU, consistent with all patient-level analyses.

5 Functional form: mortality ceiling

Fitting $P(\text{mortality}) = m_{\max} (1 - e^{-k \cdot \text{FE rate}})$ at the caregiver level ($N = 90$ caregivers with FE rate > 0). This is reparametrized from the equivalent survival floor form: $m_{\max} = 1 - b$ where b is the survival floor. The mortality ceiling $m_{\max}$ represents the maximum achievable 12-month mortality as FE rate $\rightarrow \infty$ — equivalently, the fraction of outcomes potentially sensitive to caregiver extubation practice.

Parameter	Estimate	95% CI
$m_{\max}$ (mortality ceiling)	0.357	0.321–0.397
k (rate of increase)	76.4	50.8–124.0

Approximately 36% of outcomes in this cohort are potentially sensitive to caregiver FE rate; 64% of patients reach their 12-month outcome regardless of caregiver practice. The CVICU-specific model gives $m_{\max} = 0.318$ (95% CI 0.263–0.398), consistent with CVICU’s overall lower mortality. The increase is steep: most of the mortality difference occurs in the 0–8% FE rate range.

6 Mixed effects models

Model	Estimate	95% CI	p / $P(>0)$
glmmTMB (caregiver_fe_rate, log-odds)	+0.460	–	$p = 0.660$
rstanarm (caregiver_fe_scaled, posterior)	+0.017	(–0.071, 0.103)	64.6%

Random effect variances: caregiver_id ≈ 0 (both models); first_careunit = 0.295 (glmmTMB), posterior mean 0.501 (rstanarm). The unit-level random effect absorbs mortality differences across units, leaving the caregiver FE rate fixed effect with no global signal to explain. This is consistent with all other analyses showing the signal (to the extent it exists) is CVICU-specific, not global. The near-zero caregiver-level random effect variance is also consistent with the low denominator R^2 values in the MSM: patients are not systematically routed to high-FE caregivers based on their covariates, and caregiver-level clustering explains little variance in outcomes once unit is accounted for.

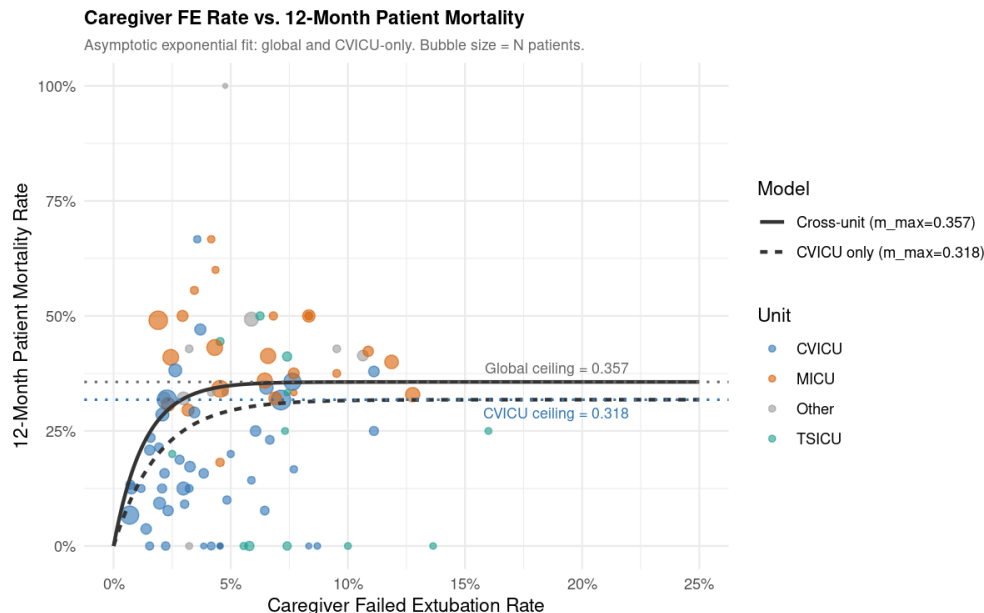


Figure 9: Caregiver FE rate vs. 12-month mortality at the caregiver level. Asymptotic exponential fits: global (cross-unit, $N = 90$) and CVICU-only ($N = 44$). Mortality ceiling $m_{\max} = 0.357$ globally, 0.318 for CVICU.

7 Deep Learning Models

A Python pipeline trains three models on the same patient-level covariates used in the R analyses (70/15/15 train/val/test split, seed 237): L2-regularized logistic regression, gradient-boosted trees (GBM), and a multilayer perceptron (MLP) with entity embeddings for ICU unit, intubation type, primary diagnosis, and sex. These are predictive benchmarks, not causal models. They ask whether flexible nonlinear interactions improve discrimination and whether caregiver FE rate contributes to prediction once the network can exploit richer feature interactions.

7.1 Predictive performance

Model	AUC-ROC	AUC-PR	Brier
Logistic regression (test)	0.829	0.663	0.149
Gradient boosting (test)	0.852	0.674	0.141
MLP (test, best ep. 16)	0.841	0.670	0.166

The MLP matches gradient boosting closely (test AUC 0.841 vs. 0.852 at early-stopping epoch 16), indicating no deep nonlinear structure beyond what the tree ensemble already captures at this sample size ($N = 3,437$). All models substantially outperform the mortality-rate baseline (AUC-PR ≈ 0.30 for a random classifier at 30% prevalence), consistent with the R regression models showing that SOFA, Charlson, and ICU unit predict outcomes well.

7.2 SHAP feature importance

The SHAP ranking (Figure 10) is consistent with the R analyses. Charlson comorbidity index is the dominant predictor by a wide margin; SOFA ranks second and `first_careunit` (ICU unit

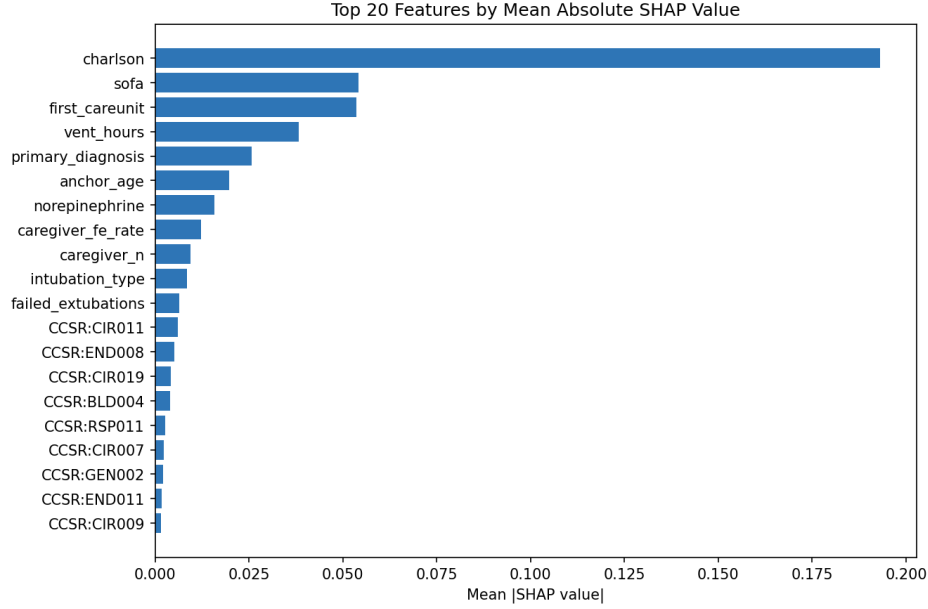


Figure 10: Top 20 features by mean $|\text{SHAP value}|$ from the MLP (200 test-set patients, KernelExplainer). Features are ordered by global importance.

embedding) ranks third — confirming that illness severity and unit-level case-mix differences are the primary drivers of mortality heterogeneity. Ventilation hours, primary diagnosis, and age follow. Caregiver FE rate ranks 8th out of 36 features, with a positive direction in the SHAP dependence plot (higher FE rate pushes predicted mortality upward), consistent with all R analyses. Its modest rank relative to Charlson and unit reflects the same finding as the mixed-effects and MSM models: FE rate carries directionally consistent signal within units but is dominated globally by patient severity and case-mix.

7.3 Double/Debiased ML (DML)

The partially linear DML estimator (Chernozhukov et al. 2018) uses 5-fold cross-fitting with GBM nuisance models to partial out all measured confounders from both the outcome and the treatment, then estimates $\hat{\theta}$ by regressing the mortality residual on the FE rate residual. The estimand is $\Delta P(\text{mortality})$ per unit of FE rate. The analysis has been rerun on the updated cohort ($\text{caregiver_unit_n} \geq 20$, outcome = 12-month mortality as defined in the Python pipeline; see note below on the mortality definition) stratified by ICU group, matching the R MSM analysis.

Treatment R^2 interpretation. The GBM treatment model achieves $R^2 \approx 0.53\text{--}0.69$ for the unit-stratified models and $R^2 \approx 0.52\text{--}0.60$ globally, substantially higher than the linear OLS denominator $R^2 = 0.057$ from the R MSM. The GBM is capturing nonlinear or interaction-based patterns in FE rate that OLS misses. The most likely explanation is caregiver-level clustering: patients of the same caregiver share the same FE rate, and patient profiles are weakly correlated with caregiver identity in ways a GBM can exploit. Despite this higher treatment predictability, all $\hat{\theta}$ estimates are null — confirming that even the additional variation the GBM explains does not translate to a causal FE rate effect on mortality.

All three groups are null. CVICU is directionally positive ($\hat{\theta} = +1.062$, IQR effect +2.3 pp,

Group	N	$\hat{\theta}$	95% CI	p	IQR effect
CVICU	1,435	+1.062	[−0.550, +2.673]	0.197	+2.3 pp
Medical	1,001	−0.542	[−1.980, +0.896]	0.460	−2.2 pp
Surgical/Trauma	809	−0.696	[−2.163, +0.771]	0.352	−2.2 pp
Global (<code>explicit_extubations</code>)	3,437	−0.333	[−1.010, +0.344]	0.335	−1.4 pp

Table 10: DML results. Unit-stratified rows use `caregiver_unit_n` ≥ 20 ($N = 3,245$ primary-group patients). Global row uses `explicit_extubations` ($N = 3,437$; `caregiver_n` ≥ 20 , all units). IQR effect = $\hat{\theta} \times$ IQR of `caregiver_fe_rate` within the group. All estimates are null. CVICU is directionally positive (higher FE rate $\rightarrow$ higher mortality), consistent with the MSM and the caregiver-level partial correlation. Medical, Surgical/Trauma, and Global are directionally negative but also null. Overall cohort mortality rate 30.1% matches the R analysis.

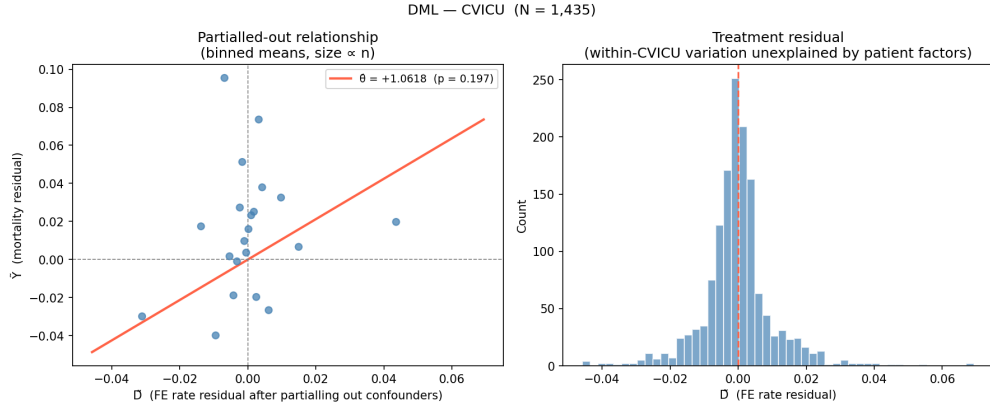


Figure 11: DML partialled-out scatter (left) and FE rate residual distribution (right) for CVICU. $\hat{\theta} = +1.062$, $p = 0.197$. The wide confidence interval reflects both the small IQR of FE rate in CVICU (2.1 pp) and the limited residual variation in FE rate after GBM partialling.

$p = 0.197$), consistent with the MSM and the caregiver-level partial correlation. Over the CVICU FE rate IQR of 2.1 pp, the implied mortality difference is just 2.3 pp — small and well within the confidence interval. Medical and Surgical/Trauma are directionally negative but also null ($p = 0.46$, 0.35). The wide confidence intervals in all groups reflect the limited residual FE rate variation after the GBM has removed patient-level confounding (treatment $R^2 \approx 0.53$ – 0.69), confirming that the causal signal, if it exists, is smaller than what these data can resolve.

8 Trajectory Analysis

Andrew’s proposed dynamic risk framework requires that reintubation risk be estimable as a function of time on the ventilator, so that the optimal extubation time can be identified as the point where one more day of waiting produces less than approximately 0.5% reduction in failure probability. We tested whether two standard readiness variables — RSBI and P/F ratio — carry the trajectory information this framework requires.

8.1 Data

RSBI (spontaneous respiratory rate / spontaneous tidal volume in liters) and P/F ratio ($\text{PaO}_2/\text{FiO}_2$) were pulled from MIMIC-IV for all explicit extubation events in the `all_extubations` cohort (6,063 events, 5,505 patients). The outcome is `failed_extubation_flag`: reintubation within 72 hours of this specific extubation event. This is the per-event outcome, not the patient-lifetime reintubation count.

Variable	Events with ≥ 3 measurements	Failed extubations
RSBI	3,551	395 (11.1%)
P/F ratio	3,747	345 (9.2%)

8.2 Individual-level slope analysis

For each extubation event with ≥ 3 measurements in the 72h window, a linear slope was estimated by fitting `lm(measurement ~ hours.before_extubation)`. A negative slope indicates the variable is improving as extubation approaches.

	Succeeded		Failed	
	Mean slope	Mean at extub	Mean slope	Mean at extub
RSBI	-0.782	48.7	-0.290	52.3
P/F ratio	-0.627	256	-0.272	251

Both variables show slopes in the expected direction — succeeded events have more steeply declining (improving) trajectories — but the difference is small relative to within-patient noise.

Logistic regression testing whether slope predicts failure beyond level at extubation:

Predictor	Coefficient	SE	z	p
<i>RSBI model:</i>				
Level at extubation	0.0032	0.0015	2.14	0.032
Slope (72h window)	0.0017	0.0036	0.48	0.634
<i>P/F ratio model:</i>				
Level at extubation	-0.0004	0.0006	-0.67	0.505
Slope (72h window)	0.0015	0.0044	0.35	0.726

RSBI level at extubation is a significant predictor ($p = 0.032$); neither slope is significant ($p = 0.634$ and $p = 0.726$). The trajectory of RSBI or P/F ratio in the 72 hours before extubation carries no independent predictive information beyond the level at the moment of extubation.

8.3 Population-level trajectories

Figure 12 shows hourly mean RSBI and P/F ratio in the 72h before extubation, split by outcome.

8.4 Interpretation

The null slope finding is consistent with two non-exclusive explanations. First, RSBI and P/F ratio may not be sensitive enough variables to detect the trajectory signal — they are measured infrequently (median 4–6 measurements per patient in the 72h window) and are inherently noisy.

Physiological Trajectories Before Extubation by Outcome

Per-extubation-event data (RSBI: 5046 events; P/F: 5327 events).

Hourly bins, $n \geq 30$ per bin. Mean $\pm$ 95% CI. Failed = reintubated within 72h of this specific extubation.

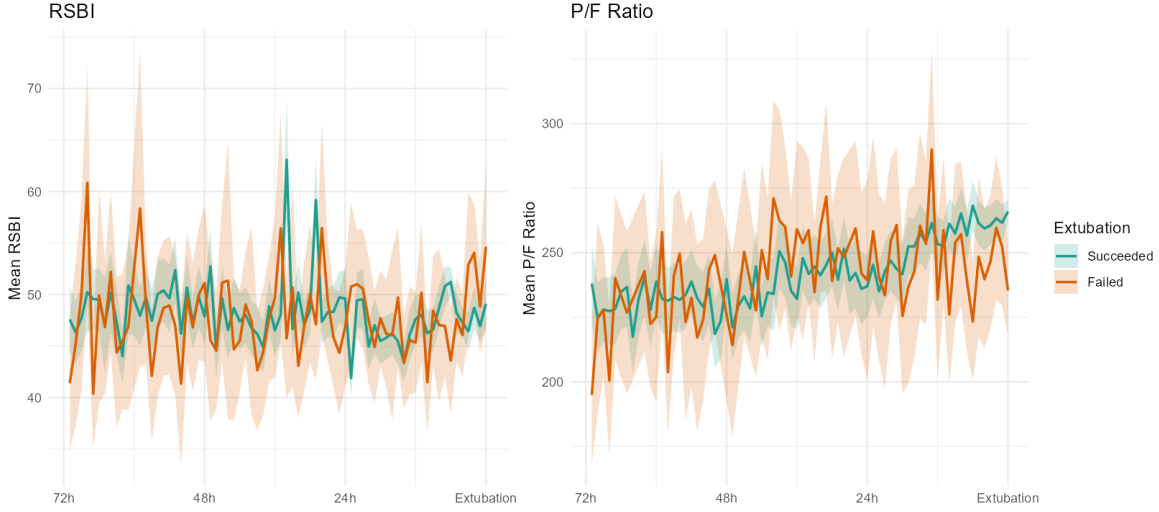


Figure 12: RSBI and P/F ratio in the 72h before extubation by outcome (failed vs. succeeded). Hourly bins, $n \geq 30$ per bin, mean $\pm$ 95% CI. The two groups are largely indistinguishable throughout the window. The zigzag in the failed group reflects genuine measurement noise with approximately 400 failed events distributed across 72 hourly bins.

Second, clinicians may already be conditioning on trajectory implicitly — extubating only after the patient has plateaued at a low RSBI — so that by the time of extubation the trajectory information is spent.

The dynamic risk framework remains theoretically sound. Implementing it prospectively would require variables with higher temporal resolution than are available in retrospective EHR data: continuous pressure support level, FiO_2 weaning trajectory, and spontaneous breathing trial (SBT) outcomes are candidates. Retrospective validation from MIMIC-IV data alone is not feasible with current measurement density.

TODO — SBT outcomes: Spontaneous breathing trial pass/fail is documented in MIMIC-IV chartevents. Whether SBT outcome predicts extubation failure beyond RSBI level is a direct test of the dynamic risk hypothesis and is feasible with existing data.

9 Limitations

1. **Caregiver attribution.** `caregiver_id` records the documenting user, not necessarily the provider who performed the extubation. This introduces measurement error in the exposure variable that would tend to attenuate any real signal toward the null.
2. **Case mix confounding within CVICU.** The volume confounding analysis demonstrates that 0%-FE-rate caregivers systematically see shorter, simpler cases. Even with `caregiver_unit_n` ≥ 20 , residual case mix differences that are not captured by the measured covariates may explain the partial correlation ($r = +0.261$) seen at the caregiver level. The null causal estimates across methods (MSM OR = 1.432, $p = 0.102$; DML $\hat{\theta} = +1.062$, $p = 0.197$) are the strongest evidence

that this association does not reflect a causal effect once patient-level confounders are removed.

3. **Single center.** MIMIC-IV is from Beth Israel Deaconess Medical Center. An eICU parallel analysis (208 ICUs, with possible caregiver type information) is planned to assess generalizability.
4. **Caregiver type unknown.** Whether `caregiver_id` corresponds to an attending physician, APP, or trainee is not directly available in MIMIC-IV. Andrew asks whether this information exists; it may be recoverable via the `caregivers` table, which requires a targeted BigQuery query.
5. **Terminal extubation misclassification.** The 3-day mortality criterion captures some patients extubated with survival intent who deteriorated unexpectedly. A code status change marker would be more specific (planned; see Section 1.3).
6. **All-cause 12-month mortality.** Deaths unrelated to the index extubation are counted. Shorter-term outcomes (hospital and 30-day mortality) are consistent in direction with the 12-month endpoint.
7. **Missed extubation opportunities.** Patients extubated too late cannot be identified in retrospective data without knowing the counterfactual. This is the motivation for the dynamic risk analysis (Section 8).

10 Proposed Next Steps

1. **Code status query (BigQuery).** Pull DNR/DNI orders from MIMIC-IV `chartevents` and identify code status changes in the 24–48h before extubation. This provides a more specific terminal extubation marker than the 3-day mortality proxy.
2. **CONSORT diagram.** Cohort flow figure (script 09_consort.R) for the paper.
3. **SBT outcomes analysis.** Spontaneous breathing trial pass/fail documented in MIMIC-IV `chartevents` (see Section 8).
4. **eICU exploratory analysis.** 208 ICUs, potential caregiver type information, external validation of CVICU partial correlation finding.
5. **Citation manager.** Zotero recommended.